MOD-0011 - noknok PicoHub with CH32X035 power manager

Power tree and X035 pin map: docs/spec.md. Placement rules: docs/layout-guide.md.

PD input and power manager

File: pd_input.kicad_sch

Power rails

File: power.kicad_sch

Net classes - required trace widths and schematic colours

Widths are the REQUIRED minimum for the rated current: IPC-2221, dT 20 C, 1 oz outer copper, with ~2x margin.
Route at the full width; necking down for the last fraction of a millimetre into a narrow pad is fine and expected.

colour	class	width	clearance	via	nets
	HV	1.00 mm	0.25 mm	0.8 / 0.4	VBUS_PD, J1 VBUS, F1, R3 shunt node - 20 V, 3 A input path
	Power3A	1.00 mm	0.20 mm	0.8 / 0.4	+5V, U2 SW node - 3 A buck output, keep the SW loop tiny
	Power	0.50 mm	0.20 mm	0.8 / 0.4	+3V3_A, +3V3_B, MEZZ_5V, USB_VBUS, PICO_VSYS, LDO inputs - up to 2 A
	USB	0.25 mm	0.20 mm	0.6 / 0.3	USB_DP, USB_DN, D6/D7 side - full speed only; tight pair, clear of the buck
	Default	0.25 mm	0.20 mm	0.6 / 0.3	logic, EN_*, MEZZ_* signals, I2C, SWD, CC, sense taps, +3V3_MCU
	GND	0.50 mm	0.20 mm	0.6 / 0.3	solid pour on B.Cu - width applies to stubs only; stitch generously

Current basis: 3 A = 0.90 mm, 2 A = 0.51 mm, 1.45 A = 0.33 mm. HV clearance is 0.25 mm because KiCad SOT-23 pads leave
only 0.35 mm between pins; IPC-2221 B2 asks 0.1 mm at 20 V. Board minimum clearance 0.15 mm. Full rules: docs/layout-guide.md.
A class width is the DEFAULT for new tracks, not a DRC rule - wider is always allowed and several rails are drawn wider.

MOD-0011 - noknok PicoHub. Generated by make_sch.ps1
noknok

Sheet: /
File: MOD-0011.kicad_sch

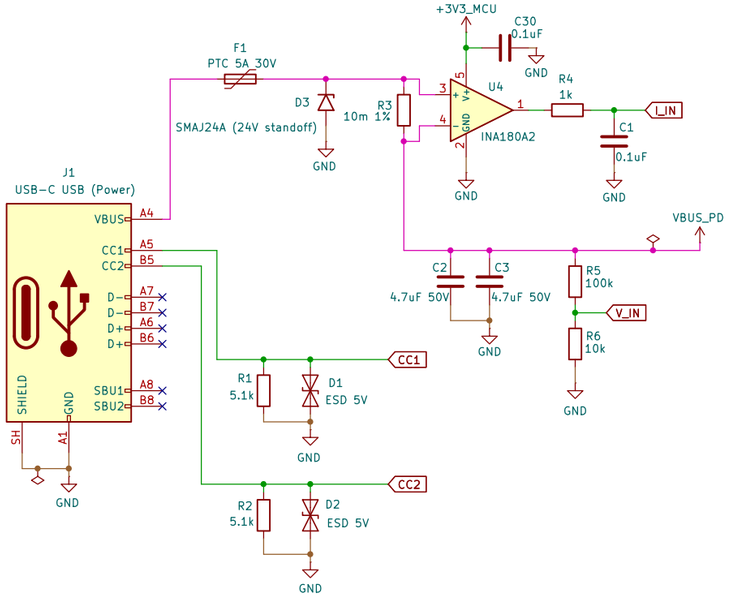
Title: MOD-0011 - noknok PicoHub

Size: A3	Date: 2026-09-21	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 1/3

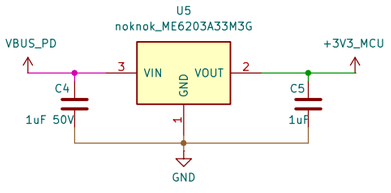
noknok Pico Hub – I2C Module
MOD-0011

Power Input & Power for MCU

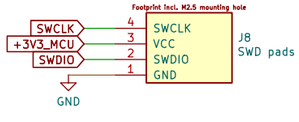
USB-C PD input: external Rd on CC, 5 A PTC, 24 V TVS,
input current (shunt + INA180) and voltage sense, sink caps <= 10 uF



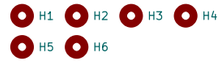
Always-on 3.3 V for the power manager:
ME6203 (40 V in) straight from VBUS_PD



SWD flashing pads (noknok 4-pad, pogo clamp)

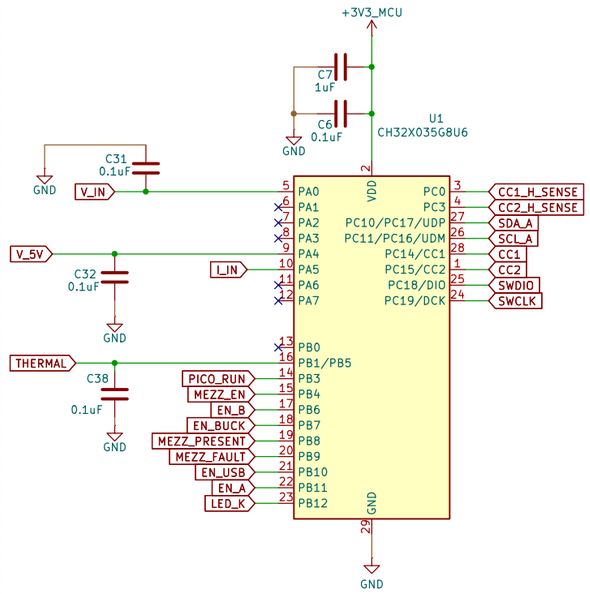


M2.5 mounting holes
shared with the mezzanine standoffs

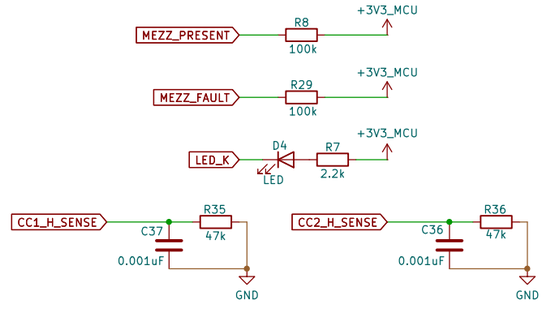


MCUs & Pull-ups

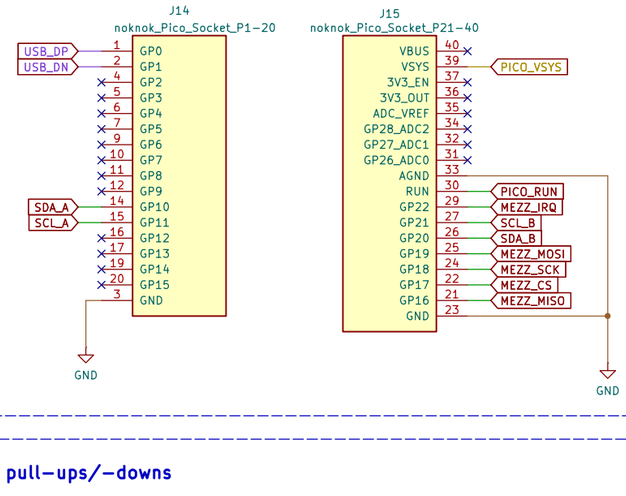
CH32X035G8U6 power manager: PD PHY on PC14/PC15,
I2C slave to the Pico on PC16/PC17 (remap 2),
2-wire SWD on PC18/PC19



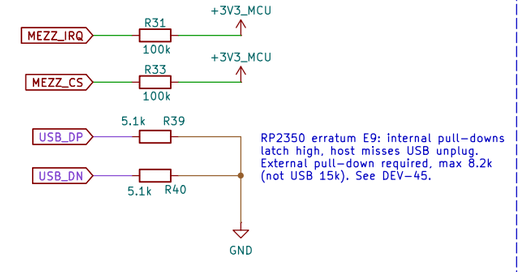
CH32X035 pull-ups/-downs and Status LED



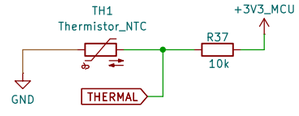
Raspberry Pi Pico / Pico 2 W on 2x20 sockets.
VSYS from +5V via SS34; micro-USB = programming only.
I2C0 GP8/9 = bus A, I2C1 GP10/11 = bus B, PIO-USB GP16/17.
SPI GP2/3/4/5, free IRQ from Mezzanine on GP6
AGND (33) is tied to GND on the Pico itself – left NC here.



Pico pull-ups/-downs



Temperature Sensor to measure TPS54302 buck heat level
Place TH1 next to GND pin of TPS54302.



noknok

Sheet: /PD input and power manager/
File: pd_input.kicad_sch

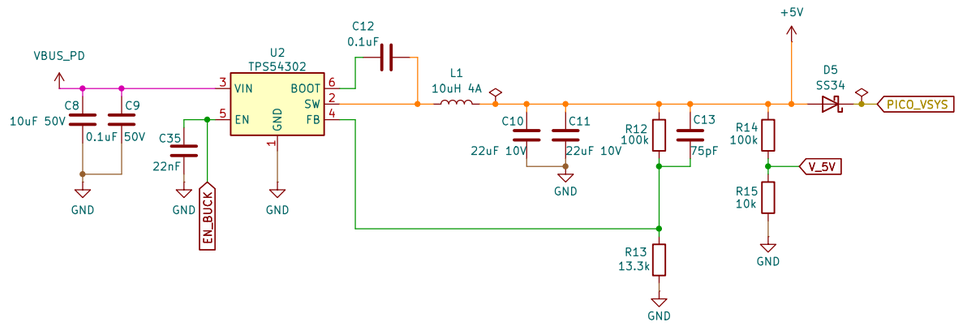
Title: MOD-0011 – PD input and power manager

Size: A3	Date: 2026-09-21	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 2/3

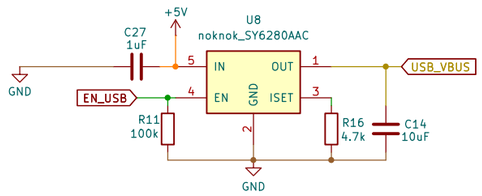
Power Rails & Power On/Off Switches

Connectors

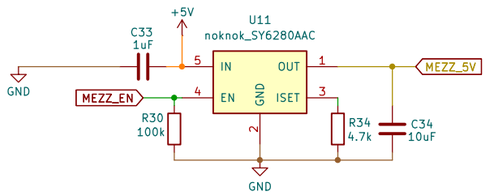
TPS54302 buck: VBUS_PD (5..20 V) -> +5V / 3 A. L 10 uH, Cout 2x22 uF,
FB 100k/13.3k (Vref 0.596 V), Cff 75 pF. EN floats = enabled.



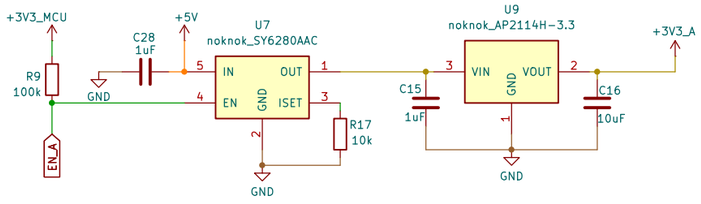
USB host port supply: SY6280 load switch,
Ilim = 6800 / 4.7k = 1.45 A,
EN pulled up = default ON



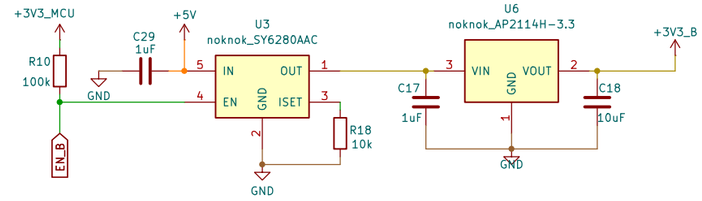
MEZZ_PRESENT low = mezzanine takes the USB uplink
→ X035 turns EN_USB off.



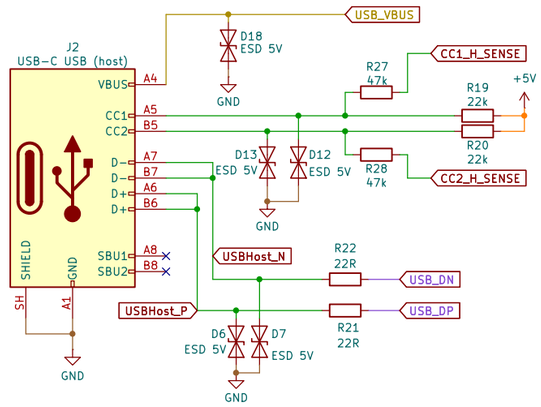
I2C bus A rail: +5V -> SY6280 (Ilim 6800 / 10k = 0.68 A, EN pulled up = default ON)
-> AP2114H-3.3 -> +3V3_A. The 0.68 A limit is the LDO thermal budget (1.15 W in SOT-223).



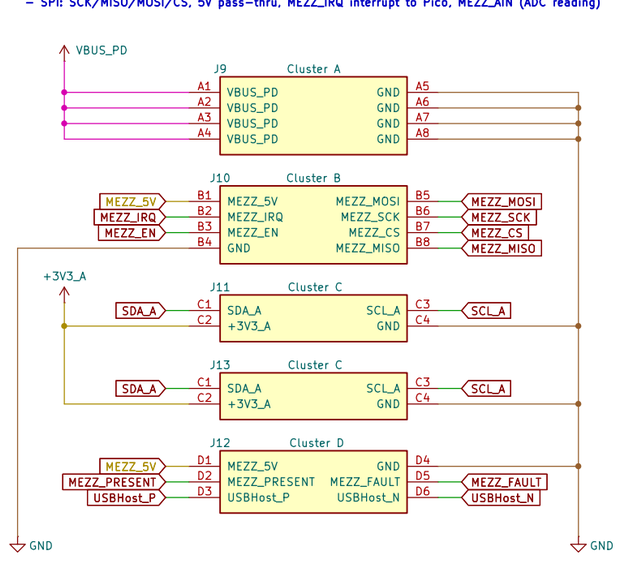
I2C bus B rail: +5V -> SY6280 (Ilim 6800 / 10k = 0.68 A, EN pulled up = default ON)
-> AP2114H-3.3 -> +3V3_B. The 0.68 A limit is the LDO thermal budget (1.15 W in SOT-223).



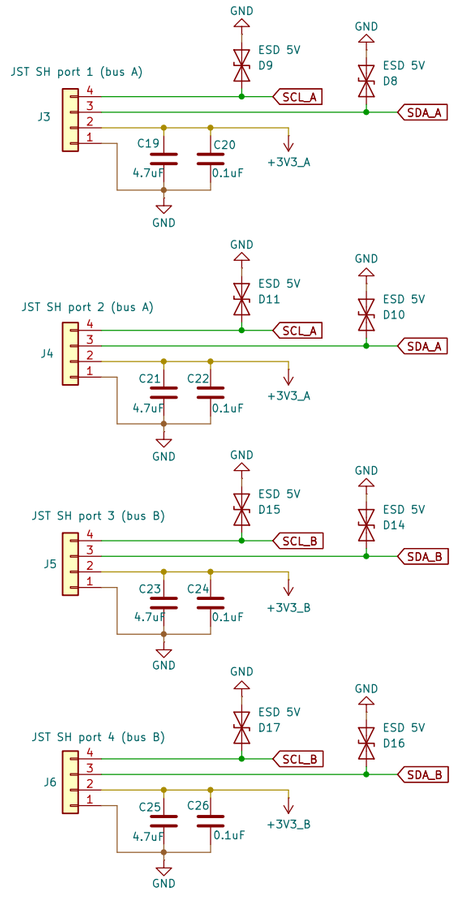
USB-C source port for one USB module:
5 V from the SY6280 (1.45 A),
Rp 22k from +5V = 1.5 A advertisement,
CC1/CC2 -> 10k -> X035 PA3/PA4:
device-present detect (VBUS off until a sink is seen),
ESD at the socket, 22 R series to the PIO-USB pins GP16/17
(also routed to the mezzanine pads)



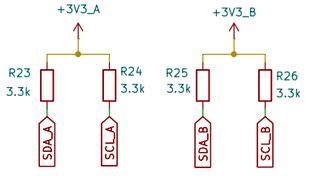
Mezzanine contact pads (bottom, under the Pico).
Grouped in 4 clusters:
- High power path: 4x VBUS_PD + 4x GND (5 A), MEZZ_EN (enabled)
- Pico USB host: D+/D-, 5V pass-thru, GND, MEZZ_PRESENT (USB used by MEZZ), MEZZ_FAULT
- I2C: SDA/SCL, 3.3V pass-thru, GND
- SPI: SCK/MISO/MOSI/CS, 5V pass-thru, MEZZ_IRQ interrupt to Pico, MEZZ_AIN (ADC reading)



Qwiic / Stemma QT ports.
Ports 1-2 = bus A on +3V3_A, 3-4 = bus B on +3V3_B.
Local 4.7u + 100n per port, ESD on SDA/SCL,
pull-ups 3.3k once per bus on the BUS rail.



Bus pull-ups (3.3k, on the bus rail)



noknok

Sheet: /Power rails/
File: power.kicad_sch

Title: MOD-0011 – Power rails

Size: A3	Date: 2026-09-21	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 3/3